

Joint PCA–QUBO margin for equities, European and American options

Version 2: implemented extension

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What changed. The original document described equities and European equity/spot-index options. The new research API accepts vanilla calls and puts of either exercise style, with long or short positions. European values use Black–Scholes; American values use the Ju–Zhong approximation specified in the supplied `american_pca_qubo_model.md`. Calibration, horizon valuation and sensitivities use the same selected pricing formula. Boundary-aware candidate generation supplements QUBO proposals, and final ranking uses full nonlinear portfolio P&L.

The existing PCA geometry and integer-ball QUBO algebra are retained. No binary exercise choice is added. With three stress coordinates and eight bits per coordinate, the QUBO still has 122 variables, regardless of option count. Multiple smooth anchors generate separate QUBOs on the same stress domain.

Scope. The supported instruments are vanilla equity and spot-index options with continuous dividend yield and European or continuous American exercise. American pricing requires nonnegative rates and yields, positive spot, strike and IV, and nonnegative remaining maturity. Rates, yield and IV are held fixed across scenarios. Futures options, discrete dividends, exotic exercise/payoffs, FX and stochastic volatility are outside this extension.

This is a frozen-book horizon revaluation. It treats each option as outstanding at the horizon; it does not simulate intervening exercise, assignment or their cash/stock ledger. A short American contract is a negative holding of the holder’s value, not a different exercise policy. Ju–Zhong is an approximation to American optimal stopping. The result is the *greatest model loss found*, not an exact American value, a certified worst-case loss, or a calibrated VaR.

Implementation map.

Component	Responsibility
<code>VanillaPriceContext</code>	Immutable pricing context, cached boundary, branch derivatives and domain checks.
<code>OptionBook</code>	Dated marks, contract units, bracketed IV calibration, supplied-IV diagnostics and nonlinear option P&L.
<code>OptionFactorStressModel</code>	Shared stock/option scenarios, residual alignment and mixed quadratic coefficients.
<code>solveOptionFactorStress</code>	Boundary seeds, smooth-anchor QUBOs, integer repair and nonlinear ranking.

The original European-only API and PDF remain available. Application YAML, existing option-market conventions and T0–T3 runner defaults are unchanged.

1. Portfolio units, dates and shared scenarios

For underlying i , let $S_{i0} > 0$ be the dated spot and w_i its signed monetary return exposure. A holding of q_i^{sh} shares has $w_i = q_i^{\text{sh}} S_{i0}$. For option o , let $i(o)$ identify its underlying, K_o its strike, T_o its expiry, n_o its signed contract count and μ_o its contract multiplier. Write $\beta_o = n_o \mu_o$ and $\phi_o = +1$ for a call, -1 for a put. Observed marks V_o^{market} are per quoted unit. The multiplier is applied exactly once, and all monetary inputs must use one currency.

With valuation date t_0 and horizon t_h , ACT/365 maturity is

$$\tau_o = \frac{\text{days}(T_o - t_0)}{365}, \quad \tau'_o = \frac{\text{days}(T_o - t_h)}{365}.$$

The option must be unexpired at t_0 . At $\tau'_o = 0$ use $[\phi_o(S - K_o)]_+$. Expiry before t_h fails because the terminal scenario does not supply settlement data. Underlying return observations must match the horizon; changing the option date gap does not rescale the fitted return model. Nominal mark-change P&L excludes funding, stock-dividend cash credits and FX.

For aligned horizon-compatible historical log returns ℓ_{ti} , oldest first, use the same normalized exponential weights in all fitted moments:

$$\omega_t = \frac{\alpha^{n-t}}{\sum_u \alpha^{n-u}}, \quad c_i = \sum_t \omega_t \ell_{ti}, \quad \sigma_i^2 = \sum_t \omega_t (\ell_{ti} - c_i)^2, \quad x_{ti} = \frac{\ell_{ti} - c_i}{\sigma_i}.$$

For nonzero variance, let $D = \text{diag}(\sigma_i)$ and $C = \sum_t \omega_t x_t x_t^\top$, with descending eigenpairs (u_a, λ_a) . The existing PCA layer owns degenerate-history validation and numerical scaling; this extension does not change that layer. Retaining p directions gives

$$\epsilon_t = D(I - U_p U_p^\top) x_t, \quad R = \sum_t \omega_t \epsilon_t \epsilon_t^\top.$$

At a smooth portfolio center, let v be the full stock/option log-return gradient. The baseline residual direction and factor matrix are

$$a_{\text{res}} = \frac{Rv}{\sqrt{v^\top Rv}}, \quad A = [Du_1 \sqrt{\lambda_1}, \dots, Du_p \sqrt{\lambda_p}, a_{\text{res}}],$$

with $a_{\text{res}} = 0$ if $v^\top Rv = 0$. Weighted observation products avoid a dense residual covariance. This preserves local linear residual variance; it can miss residual gamma risk, particularly in delta-neutral option books.

All positions on underlying i share

$$\ell(z) = c + Az, \quad S_i(z) = S_{i0} e^{c_i + a_i^\top z} = \bar{S}_i e^{a_i^\top z}, \quad \|z\|_2 \leq \rho.$$

The configured radius is a stress budget. PCA alone does not give it a probability interpretation. At $z = 0$, log return is c , not zero.

2. European and American valuation

For positive remaining maturity τ , volatility ν , rate r and continuous yield q , define

$$d_1 = \frac{\log(S/K) + (r - q + \nu^2/2)\tau}{\nu\sqrt{\tau}}, \quad d_2 = d_1 - \nu\sqrt{\tau},$$

$$V_E = \phi[Se^{-q\tau}\Phi(\phi d_1) - Ke^{-r\tau}\Phi(\phi d_2)], \quad \Delta_E = \phi e^{-q\tau}\Phi(\phi d_1), \quad \Gamma_E = \frac{e^{-q\tau}\varphi(d_1)}{S\nu\sqrt{\tau}}.$$

European spot gamma is positive for both calls and puts. The maturity derivative at fixed spot and parameters is

$$T_E = \frac{Se^{-q\tau}\varphi(d_1)\nu}{2\sqrt{\tau}} - \phi q Se^{-q\tau}\Phi(\phi d_1) + \phi r Ke^{-r\tau}\Phi(\phi d_2).$$

Ju–Zhong context. For an American option, first use the European-equivalent shortcuts: $q = 0$ for a call, or $r = 0$ for a put, under the nonnegative rate/yield restrictions. Otherwise define

$$\beta_J = \frac{2(r - q)}{\nu^2}, \quad \kappa_J = \begin{cases} 2r/[\nu^2(1 - e^{-r\tau})], & r > 0, \\ 2/(\nu^2\tau), & r = 0, \end{cases}$$

$$\zeta_J = \sqrt{(\beta_J - 1)^2 + 4\kappa_J}, \quad \eta_J = \frac{-(\beta_J - 1) + \phi\zeta_J}{2}, \quad d_J = \phi\zeta_J.$$

The implementation uses `expm1` for $1 - e^{-r\tau}$ and a stable quadratic root computation. The finite boundary S^* solves

$$U^* = \phi(S^* - K) - V_E(S^*), \quad \Delta_E(S^*) + \eta_J U^*/S^* - \phi = 0.$$

A sign-changing log-spot bracket and safeguarded scalar root solver determine the boundary. Its residual and positive premium are checked before computing

$$b_J = -\frac{e^{-r\tau}\kappa_J^2}{2\zeta_J^2}, \quad c_J = -\frac{1}{d_J} \left[\frac{2T_E(S^*)}{\nu^2 U^*} + e^{-r\tau}\kappa_J + 2b_J \right].$$

For $X = \log(S/S^*)$, set $\mathcal{D} = 1 - b_J X^2 - c_J X$ and $\mathcal{E} = U^* e^{\eta_J X} / \mathcal{D}$. The complete price is

$$\hat{V}_J(S) = \begin{cases} V_E(S) + \mathcal{E}(X), & \phi(S^* - S) > 0, \\ \phi(S - K), & \phi(S^* - S) \leq 0. \end{cases}$$

The context is cached per contract and remaining maturity. Spot-only scenarios do not recalibrate IV or solve another boundary. The final rational/exponential pricing formula is not itself a quadratic polynomial.

3. Calibration, derivatives and pricing-domain checks

Mark inversion. The new book solves $V(\nu_o) = V_{o0}^{\text{market}}$ using the contract's exercise style. Brent's method maintains a bracket; every trial American IV rebuilds the boundary. The default explicit bracket is $[0.01, 5]$. Invalid trial contexts and missing brackets fail. The final quoted-price residual must satisfy

$$|V(\nu_o) - V_{o0}^{\text{market}}| \leq 10^{-10} \max(1, V_{o0}^{\text{market}}).$$

Necessary economic mark bounds are checked, including the zero-volatility American stopping lower limit $\max_{0 \leq u \leq \tau} [\phi(Se^{-qu} - Ke^{-ru})]_+$. An American intrinsic-value plateau does not identify a unique positive IV. Such a mark requires an explicitly supplied IV; supplied-IV status and its calibration residual are recorded. Observed marks remain the P&L anchor. An American bracketed root is not a uniqueness certificate.

Derivatives of the selected formula. Write $x = \log S$, $u_J = c_J + 2b_J X$ and $L_J = \eta_J + u_J/\mathcal{D}$. In continuation,

$$\begin{aligned} D^J &:= \widehat{V}_{J,x} = S\Delta_E + \mathcal{E}L_J, \\ G^J &:= \widehat{V}_{J,xx} = S\Delta_E + S^2\Gamma_E + \mathcal{E} \left[L_J^2 + \frac{2b_J}{\mathcal{D}} + \frac{u_J^2}{\mathcal{D}^2} \right]. \end{aligned}$$

The European terms use the current spot. In the strict exercise region, $D^J = G^J = \phi S$. At expiry, these derivatives apply inside the money, and both are zero outside the money. Two-sided derivatives at the strike are undefined.

The final Ju-Zhong formula matches value at S^* but generally has a slope jump:

$$\Delta_J^{\text{continuation}}(S^*) - \Delta_J^{\text{exercise}}(S^*) = \frac{c_J U^*}{S^*}.$$

The implementation raises for requested derivatives within log-distance 10^{-10} of the boundary; it does not invent a two-sided Hessian. Price-only candidate evaluation remains valid.

Domain checks. Each contract's marginal stress interval is

$$[S_i^{\min}, S_i^{\max}] = [\bar{S}_i e^{-\rho \|a_i\|}, \bar{S}_i e^{\rho \|a_i\|}].$$

On its continuation portion, the exact quadratic denominator minimum is tested at endpoints and the vertex. A value $\mathcal{D} \leq 10^{-10}$ fails. Economic bounds are sampled at 65 log-spaced points and checked at every evaluated scenario: $\max(I, V_E) \leq \widehat{V}_J$, call value $\leq S$, put value $\leq K$, with materiality tolerance $10^{-7} \max(S, K)$. These sampled checks are not an interval-wide certificate. Nonfinite prices, invalid premiums and material bounds violations fail explicitly. No clipping or scenario-dependent backend switch is used; no automatic tree fallback is claimed.

4. Mixed P&L, residual alignment and anchor polynomials

Use $V_o = V_E$ for European contracts and $V_o = \hat{V}_J$ for American contracts, with the specified payoff/equivalent branches. Then

$$P(z) = \sum_i w_i \expm1(c_i + a_i^\top z) + \sum_o \beta_o [V_o(S_{i(o)}(z), \tau'_o) - V_{o0}^{\text{market}}].$$

This includes horizon time decay in both full P&L and the Taylor constant. For a smooth center, aggregate by underlying

$$v_i = w_i e^{c_i} + \sum_{o:i(o)=i} \beta_o V_{o,x}(\bar{S}_i), \quad h_i = w_i e^{c_i} + \sum_{o:i(o)=i} \beta_o V_{o,xx}(\bar{S}_i).$$

The full v , including American and European log deltas, selects the residual direction. The center polynomial is

$$F(z) = p_0 + g^\top z + \frac{1}{2} z^\top H z, \quad p_0 = P(0), \quad g = A^\top v, \quad H = A^\top \text{diag}(h) A.$$

The Hessian is symmetrized for floating-point roundoff. Signed positions and all legs on one underlying are aggregated before optimization; there is no independent worst scenario for each option.

At a nonsmooth center, default PCA construction fails instead of manufacturing Greeks. An explicit `localPnlGradient` may select alternative residual alignment, or the caller may supply fixed geometry directly. The diagnostic distinguishes mixed-delta, supplied and fixed-geometry construction. This choice does not make the nonsmooth center differentiable. Price-only seeds and other smooth anchors remain available. The optional extra negative-curvature residual mode proposed in the supplied Markdown is not enabled by this implementation.

Reachable boundaries. For an American boundary $B_o = S_o^*$, or expiry strike $B_o = K_o$, put $b_o^* = \log(B_o/\bar{S}_{i(o)})$. The boundary plane is

$$a_{i(o)}^\top z = b_o^*.$$

For a nonzero loading row, it intersects the ball exactly when $|b_o^*| \leq \rho \|a_{i(o)}\|$, and its nearest point is $z_o^* = b_o^* a_{i(o)} / \|a_{i(o)}\|^2$. Zero loading is handled separately without division. A boundary touching the ball or a coarse lattice need not have feasible points on both sides.

The search generates center and axis seeds, nearby boundary-side seeds and marginal extreme seeds. After integer rounding/projection it checks which boundary sides are actually represented. Extreme seeds help expose the loss of a short expiry option whose center lies on a flat out-of-the-money payoff.

At each smooth lattice anchor z_j , form

$$F_j(z) = P(z_j) + g_j^\top (z - z_j) + \frac{1}{2} (z - z_j)^\top H_j (z - z_j).$$

In global coordinates its coefficients are $\tilde{g}_j = g_j - H_j z_j$ and $\tilde{p}_j = P(z_j) - g_j^\top z_j + \frac{1}{2} z_j^\top H_j z_j$. Keep A, c, ρ fixed. Nonsmooth anchors are fully repriced but do not generate derivative QUBOs. Multi-anchor global-ball proposals are heuristic; the optional local trust-ball encoding from the Markdown is not included.

5. Unchanged integer encoding and nonlinear selection

For $k \in \{2, \dots, 8\}$ bits per coordinate, write

$$m = 2^{k-1} - 1, \quad t_a = \sum_{j=0}^{k-1} 2^j b_{aj} - m, \quad z_a = \frac{\rho}{m} t_a, \quad \sum_a t_a^2 \leq m^2.$$

The extra representable unsigned level is excluded by the exact ball budget. For product bits $y_{aij} = b_{ai}b_{aj}$, the squared integer norm is represented by the linear expression

$$Q(b, y) = dm^2 + \sum_{a,j} (2^{2j} - 2m2^j) b_{aj} + 2 \sum_a \sum_{i < j} 2^{i+j} y_{aij}.$$

Let $s = \sum_{h=0}^{L-1} 2^h u_h$ with $L = \lceil \log_2(m^2 + 1) \rceil$. For the actual anchor polynomial F_j , the QUBO is

$$\begin{aligned} E(b, y, u) = & F_j(z(b)) + \mathcal{P}[Q(b, y) + s - m^2]^2 \\ & + \mathcal{P} \sum_a \sum_{i < j} [b_{ai}b_{aj} - 2b_{ai}y_{aij} - 2b_{aj}y_{aij} + 3y_{aij}]. \end{aligned}$$

Each product penalty is nonnegative and vanishes exactly for a consistent product. An invalid integer encoding incurs an unweighted penalty of at least one. If B is the sum of absolute nonconstant binary objective coefficients, the default $\mathcal{P} = 1.1 \max(B, \varepsilon)$ is sufficient for global feasibility in exact arithmetic. Each anchor recomputes B for its polynomial. Weaker configured penalty multipliers may remove that guarantee.

The number of variables remains

$$N = dk + d \binom{k}{2} + L.$$

For $d = 3, k = 8$, this is $24 + 84 + 14 = 122$. Option count increases pricing and anchor-search cost, but not the variable count of each QUBO.

Repair and ranking. Every raw solver output receives integer/product/slack diagnostics. The new search projects its integer coordinates into the ball and rebuilds auxiliaries before pricing; an invalid raw sample can lie outside the screened domain. Deterministic seed candidates are also repaired. Nonlinear local improvement prices only feasible neighbors and accepts strict decreases in complete P . The original 26-neighbor mode applies at three dimensions, with explicit pairwise repair above three dimensions. Convergence versus the move cap is retained for every candidate.

For all retained candidates $\mathcal{C}$, the reported result is

$$M_{\text{found}} = \max_{z \in \mathcal{C}} \max(0, -P(z)).$$

The result includes the worst coordinates and stressed spots; stock, European and American P&L; repaired feasible encodings; raw feasibility; anchor counts; and lattice boundary-side coverage. Calibration and pricing-domain diagnostics are retained by the book/model and archived by the runnable example. The existing solver interface returns one selected binary sample per solve; this search ranks those returned samples and its deterministic seeds after repair, not every internal solver trajectory.

No stock-only convexity certificate applies to a nonzero option position. Neither QUBO optimality nor neighborhood convergence establishes a global minimum of the nonlinear American portfolio value.

6. Validation, reproducibility and interpretation

The implemented formula reproduces these supplied numerical examples:

Inputs ($S = K = 100$)	Ju–Zhong price
Call: $\tau = 0.5, r = 0.03, q = 0.07, \nu = 0.2$	4.76818024
Put: $\tau = 3, r = 0.08, q = 0, \nu = 0.2$	6.95561233

For the put, the implementation obtains

$$S^* = 82.2759889300, \quad U^* = 8.3733727010, \quad \Delta_J^{\text{cont}} - \Delta_J^{\text{exercise}} = 0.03133987185.$$

Independent log-price finite differences agree with analytic first/second derivatives on smooth continuation and exercise branches. Existing independent CRR trees at 800 and 1,600 steps refine to within 0.01 quoted units for these two cases; the Ju–Zhong/reference difference remains visible. This validates implementation examples, not a uniform American-pricing error bound.

The focused suites passed 48 tests. The complete Python suite ran 363 tests successfully with 9 skips. Checks include mixed-style calibration, explicit-IV residuals, zero-rate limits, immutable contexts, signed multipliers, calendar maturity, mixed gradients/Hessians, residual alignment, exact QUBO energy, fixed variable counts, nonsmooth anchors, expiry plateau escape, boundary-side coverage and feasible-only repair. Native and GPU solver kernels are unchanged; no separate GPU benchmark or native rebuild was needed for this CPU extension.

Run the synthetic end-to-end example from the repository root:

```
PYTHONPATH=src python tools/example_option_factor_stress.py
```

To archive it, add `-output` followed by `docs/benchmarks/american_factor_stress_example.json`. It fits underlying PCA, creates European/American call and put marks, calibrates IV, solves the anchor QUBOs with simulated annealing, repairs and selects the greatest model loss. Its JSON separates same-IV CRR prices at the final worst scenario from the model margin. It is an offline composition demonstration, not a market-data coverage study or an exercise/assignment backtest.

Separate approximation limits. Residual reduction, finite lattice resolution, Taylor proposals, finite solver search and Ju–Zhong pricing each introduce different limitations. Nonlinear repricing removes Taylor error from the reported value of a retained candidate; it does not recover unsearched scenarios or remove the American-pricing approximation. The additional curvature residual and local trust-ball designs in the supplied Markdown remain optional future extensions. Domain failures are explicit; a material unpriced contract is never assigned zero risk.

Sources and implementation notes. The primary specification is the supplied `american_pca_qubo_model.md`, dated 12 September 2026. Its reported 87-case paper-table audit was not rerun here; only the implementation tests described above are claimed. The cited Ju–Zhong paper is N. Ju and R. Zhong, “An Approximate Formula for Pricing American Options,” *The Journal of Derivatives* 7(2), 1999, 31–40. The stable coefficients and corrected European Greek terms were cross-checked against the official QuantLib Ju engine source; QuantLib is not a dependency. Further details and commands are in `docs/benchmarks/american_factor_stress.md`.